

Capstone Project: Advanced Business Intelligence Dashboard in Excel

folder Exercise Files

Part 1: Data Preparation & Cleaning (1.5 hrs)

1. Consolidate Sales Data

- o Combine Sales_Q1, Sales_Q2, Sales_Q3 into one Sales_All sheet.
- o Ensure all dates are in UK format (DD/MM/YYYY).
- o Remove duplicate orders.

2. Customer Data Cleanup

- o Standardise names (use PROPER()).
- o Extract **First Name, Last Name, and Email domain** (RIGHT, FIND).
- o Identify and flag duplicate customers with conditional formatting.

3. Product Data Cleanup

- o Merge Product_List with Sales_All using **VLOOKUP** or **XLOOKUP**.
- o Bring in Product Category, Unit Cost, and Discount %.

4. Finance & Operations Data

- o Link Finance_Data (Overheads & Salaries) to calculate **Net Profit**.
- o Link Operations_Data (Delivery Time, Returns) to measure **Operational Efficiency**.

Part 2: Advanced Calculations & Functions (2 hrs)

1. Create **calculated fields** in Sales_All:

- o Revenue = Quantity * Unit Price
- o Discounted Revenue = Revenue * (1 – Discount%)
- o Profit = Revenue – (Quantity * Unit Cost) – Allocated Overheads

2. Use **Nested IFs / IFS** to classify customers:

- o VIP: Revenue > 50,000
- o Regular: 20,000–50,000
- o Low Value: < 20,000

3. Use **INDEX-MATCH** (not just VLOOKUP) to retrieve product details.
4. Create a **Dynamic Named Range** for Product Categories using OFFSET.
5. Apply **Array Formulas / Dynamic Arrays**:
 - o Extract unique list of Regions (UNIQUE())
 - o Count orders per Region (COUNTIF())
 - o Average delivery time per Region (AVERAGEIF())
6. Use **TEXTJOIN** with criteria to display all products purchased by a single customer.

Part 3: Data Management & Validation (45 mins)

1. Create **Data Validation Drop-downs** for Region and Product Category.
2. Add **Dependent Drop-down Lists** (e.g., selecting “Electronics” shows only electronics products).
3. Protect sheets with **locked formulas** but keep input cells open.
4. Use **Remove Duplicates + Advanced Filter** to prepare a unique “VIP Customers” report.

Part 4: Conditional Formatting (30 mins)

1. Highlight Top 10 Customers by Revenue.
2. Use a **Formula Rule**:
 - o Highlight customers whose **Revenue > 25,000 AND Returns > 5%**.
3. Apply **Icon Sets** for Delivery Performance (Green = On Time, Red = Late).
4. Apply a **3-Colour Scale** to Expenses (Low–Medium–High).

Part 5: Advanced PivotTables & Analysis (1.5 hrs)

1. Build PivotTables for:
 - o Revenue & Profit by Region → with slicers
 - o Product Category vs Net Profit → with conditional formatting
 - o Customer Classification (VIP, Regular, Low Value)
 - o Delivery Performance → Average Delivery Days by Region

2. Insert **Calculated Fields** inside PivotTables for:
 - o Profit Margin %
 - o Return Rate %
3. Add **Slicers & Timelines** for Month, Region, and Category.
4. Show **Top 5 Products by Revenue** with a dynamic filter.

Part 6: Charts & Dashboard (2 hrs)

1. Create the following charts:
 - o **Combo Chart:** Revenue (Columns) + Profit Margin % (Line)
 - o **Map Chart (if available):** Sales by Region
 - o **Stacked Bar Chart:** Expenses vs Net Profit
 - o **Pareto Chart:** 80/20 rule for Products (Top 20% of products contributing to 80% of revenue)
2. Build a **Dashboard Sheet** with:
 - o KPIs (Total Revenue, Profit, Top Customer, On-Time Delivery %)
 - o Interactive slicers & timelines
 - o Clean, professional formatting
3. Add **Dynamic Headings** (e.g., “Dashboard for [Selected Region]”) using cell references linked to slicers.

Part 7: Automation & Extras (Optional – 1 hr)

1. Use **Macros (VBA)** to:
 - o Refresh all PivotTables & Charts
 - o Export Dashboard as PDF
2. Use **What-If Analysis (Data Tables)** to simulate changes in Discount %.
3. Apply **Goal Seek**:
 - o What sales quantity is required to reach R1,000,000 profit?